*Evolution*, 2026, **0**(**0**), 1–16
https://doi.org/10.1093/evolut/qpag159
Advance access publication 11 September 2026
**Original Article**

# Repeated evolution of high-elevation occupation in tropical lizards

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## Abstract

Since Darwin and Humboldt, researchers have sought to understand how species adapt to elevation. Although tropical reptiles are typically most diverse in warm lowlands, many lineages have occupied highlands, but how often remains unclear: high-elevation species across multiple mountain ranges could reflect a single occupation event followed by diversification, or repeated occupation events with limited diversification. We tested these alternatives in anole lizards, using climatic and distributional data for 303 species and cold tolerance ($CT_{min}$) data for 64 species. Highlands have been occupied many times, though considerably less often than the reverse transition, and both dispersal among mountain ranges and climatic niche shifts are distributed nonrandomly across the phylogeny. Lowland-highland transitions did not coincide with shifts in the rate of trait evolution for either cold tolerance or minimum thermal niche. Instead, highland lineages showed stronger selection toward a colder optimum for minimum thermal niche, consistent with climatic niche conservatism in montane environments. Cold tolerance showed no such shift, indicating a decoupling between the climate lineages experience and the evolution of a physiological trait expected to track it. Overall, repeated mountain occupation has reshaped climatic niche evolution in *Anolis*, whereas cold tolerance has followed a comparatively stable, elevation-independent trajectory.

**Keywords:** macroecology, macroevolution, *Anolis*, Neotropics, cold temperatures, mountains, mountain gradient

## Introduction

Mountain ranges harbor most of the global biodiversity hotspots (Sonne & Rahbek, 2024) despite covering only a quarter of the Earth's land surface (Kapos et al., 2000). Mountain ranges can shape ecological and evolutionary patterns, making mountains useful for studying adaptation and evolution (Esquerré et al., 2018; Ibargüengoytía et al., 2021; Kapos et al., 2000; Spehn et al., 2011). In this regard, montane environments are known to facilitate the diversification of different groups of organisms, including frogs (Carvalho et al., 2021; Scherz et al., 2023; Vijayakumar et al., 2016), reptiles (Everson et al., 2020; Scherz et al., 2023), birds (Fjeldså et al., 2012; Quintero & Jetz, 2018), insects (Condamine et al., 2013; Tojo et al., 2017), and mammals (Escalante et al., 2004; Everson et al., 2020).

In ectotherms, temperature is a crucial factor that affects physiology, behavior, and distribution (Cowles & Bogert, 1944; Ghalambor et al., 2006; Huey, 1982; Janzen, 1967). Ectothermic species in temperate regions are more often exposed to cold temperatures than their tropical counterparts (Ghalambor et al., 2006; Graham et al., 2000; Janzen, 1967; Leather et al., 1993). However, ectotherms inhabiting tropical montane environments experience cold temperatures at high elevation (Ghalambor et al., 2006; Janzen, 1967).

Adaptation to montane environments requires changes in species' physiological tolerance to adjust to lower temperatures at higher elevation (Jankowski et al., 2013). In invertebrates, researchers have found that tropical insects show divergent thermal tolerance and divergent dispersal capacity: some species have broad thermal tolerance ranges while others do not, and some have low dispersal capacities while others have higher ones. This variability has led to population divergence in insect species, allowing them to evolve thermophysiological differences within short time spans (Polato et al., 2018). Similarly, lizards have also exhibited the capacity for rapid evolutionary change in cold tolerance (Gunderson & Stillman, 2015).

One group of animals that has surmounted the physiological challenges of life at high elevations in the Neotropics is the lizard genus *Anolis* (anoles). Despite the wide use of anoles to answer questions regarding the evolution of morphological and physiological traits (e.g., Ballinger et al., 1970; Calderón-Espinosa et al., 2026; Conover et al., 2015; Harmon et al., 2003; Hertz, 1974; Hertz et al., 2013; Huey et al., 2003, 2009; Losos, 2009; Mahler et

Received July 10, 2024; revisions received August 6, 2026; accepted August 11, 2026
Associate Editor: Diogo Provete; Handling Editor: Hélène Morlon
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al., 2010; Méndez-Galeano & Calderón-Espinosa, 2017; Moreno-Arias & Méndez-Galeano, 2026; Muñoz et al., 2014, 2023; Pinto et al., 2008; Ruibal, 1960; Velasco et al., 2016, 2018, 2026), fewer studies have examined how anoles have colonized mountains throughout most of the Neotropics and how these historical events may have shaped traits related to their climatic niche (e.g., Hertz, 1981; Hertz & Huey, 1981; Muñoz et al., 2014; Rodríguez et al., 2017; Salazar et al., 2019, 2025; van Berkum, 1986). Prior results indicate that phylogenetically distantly related anoles in different areas have adopted alternative strategies for adapting to novel cool montane environments. For example, anoles from the Dominican Republic and Colombia differ in their thermal physiology related to diurnal activity at high elevations. In the Dominican Republic, heat tolerance does not change as elevation increases, whereas in Colombia, heat tolerance decreases with elevation (Muñoz et al., 2014; Salazar et al., 2025). These alternative adaptive responses to elevation are the result of different thermoregulatory strategies. In the Dominican Republic, anoles are thermoregulators that, by basking more in cooler climates, are able to attain similar body temperatures across a broad elevational range (Muñoz et al., 2014). In contrast, Colombian anoles are thermoconformers that display decreasing body temperatures with increasing elevation (Salazar et al., 2025). This discrepancy highlights the challenges in understanding the evolutionary patterns of cold climate adaptation, especially for species distributed across large biogeographical regions.

The widespread occurrence of high-elevation tropical anole species raises the question of whether climatic niches and physiology are evolutionarily labile or conserved. Specifically, are high-elevation anole species across the Neotropics closely related, i.e., the result of one or a few evolutionary transitions from ancestral, low-elevation, warm-adapted ancestors? Alternatively, have there been multiple colonization events resulting in repeated adaptation to cooler environments? Here we address two hypotheses: (1) Adaptation to high-elevation environments has evolved repeatedly and independently across the anole radiation, and (2) species occupying highland environments exhibit higher rates of ecological and phenotypic evolution compared to their lowland counterparts, potentially reflecting stronger selective pressures, greater ecological opportunity in montane habitats, or greater evolvability among those lineages that are able to colonize high elevations (Table 1). Specifically, we tested whether the minimal realized thermal niche and critical thermal minimum ($CT_{min}$ or cold tolerance) elicit different evolutionary dynamics between highland and lowland lineages. Rather than inferring causality between biogeographic transitions (from lowland to highland zones and vice versa) and these thermal traits, our goal was to test for consistent macroevolutionary patterns associated with ecological regimes.

## Methods

### Phylogenetic and georeferenced data

To analyze our data, we used a time-calibrated phylogeny for 303 species based on molecular data (maximum clade credibility tree [MCC-tree]; Poe et al., 2017). We used 24,360 georeferenced locality records for 303 *Anolis* species from Caribbean and Pacific islands (145 species; Velasco et al., 2020 publicly available at https://doi.org/10.6084/m9.figshare.11778399.v1), and the mainland (158 species; Poe, 2025; Velasco et al., 2020). We extracted elevation data and environmental layers for each of the georeferenced individuals using the WorldClim2 database at a spatial resolution of 1 $km^2$ (Fick & Hijmans, 2017; Table S1). We used the thermal averages, extremes, and ranges in our analyses (see below). Additionally, we used publicly available cold tolerance ($CT_{min}$—the lowest temperature at which an individual loses motor function; Angilletta 2009) data (Bodensteiner et al., 2024; Salazar et al., 2019), which we complemented with $CT_{min}$ values from studies published between 2019 and 2025. In total, our dataset includes $CT_{min}$ data for 64 species (33 mainland and 31 island species; Table S2).

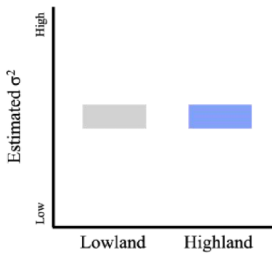
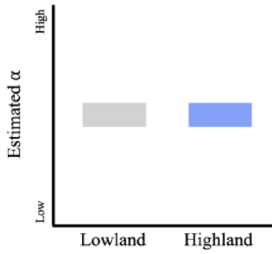
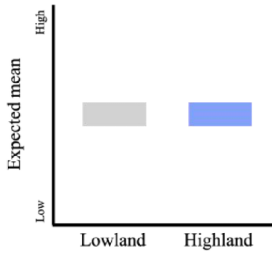
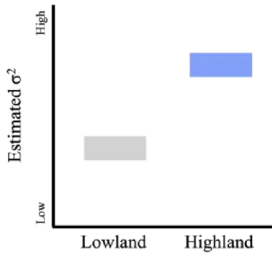
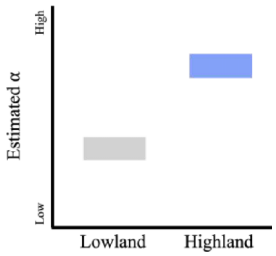
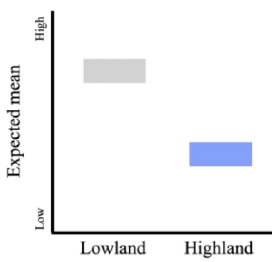
### Environmental, elevation and mountain data

Using the occurrence records of individual anoles as a proxy for the general environmental conditions experienced by a species, we extracted the minimum temperature of the coldest month (Bio 6; Table S1) from the WorldClim database (Fick & Hijmans, 2017) and calculated the mean for each species across its distribution as a measure of the minimal thermal niche -$_{min}$TN- (Frishkoff et al. 2015; Qu & Wiens, 2020), i.e., the species characteristic exposure to cold-season temperature extremes. We also extracted the elevation for each individual occurrence record using the *extract()* function from the *terra* package in R (Hijmans, 2025). Additionally, we extracted elevation data from species occurrence records and used the lowest recorded elevation for each species to assign it to an elevational category. Species with minimum elevation values above 300 (300 m criterion), 700 (700 m criterion), 1,400 (1,400 m criterion), and 2,100 m (2,100 m criterion) were classified as highland species under different thresholds.

We considered different thresholds for classifying anole species mountain-dwelling or non-mountain-dwelling (lowland-dwelling) because no consensus definition of "mountain" or "mountain areas" exists, and the threshold for an area to be considered montane is even less clear (Kapos et al., 2000). Here, we used four different thresholds: 700 m, the elevation corresponding to most elevational turnover among Caribbean anoles (Frishkoff et al., 2022; hereafter, "700 m criterion"); 300 m (hereafter referred to as the "300 m" criterion) based on geomorphological standards, as landforms rising 300 m or more above their surroundings are typically classified as mountains by geologists (Körner et al., 2016); 1,500 m (hereafter the "1,500 m criterion") that reflects well-documented shifts in vegetation structure and species composition that commonly occur between 1,500 and 1,650 m (e.g., Hu et al. 2018; Irungbam et al. 2025) since none of the island species in the analysis have a minimum elevation above 1,500 m, we decided to lower this threshold to 1,400 m; 2,100 m corresponding to the elevation at which high-mountain lizards are typically found in the Andes. (This is slightly more restrictive than the 2,000 m threshold proposed by Montoya-Cruz et al., (2024). We ended up with 70 highland and 233 lowland species at the 700 m criterion, 95 highland and 208 lowland species at 300 m, 19 highland and 284 lowland species at 1,400 m, and one highland and 302 lowland species at 2,100 m.

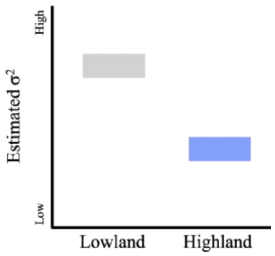
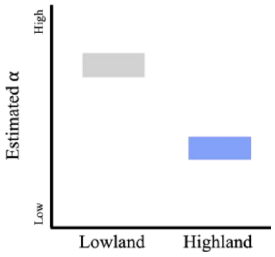
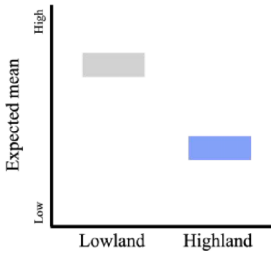
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**Table 1.** Overview of hypotheses tested using the *hOUwie* model to examine differences in OU model parameters between lowland and highland species. For each hypothesis, we present the parameter, the underlying mechanism, a specific prediction, and a visual representation of the expected relationships. Parameters: $\sigma^2$ = evolutionary rate; $\theta$ = optimal trait value; $\alpha$ = strength of selection (pull toward optimum).

| Hypotheses | Parameter | Mechanism | References | Predictions | Expected relations |
|---|---|---|---|---|---|
| Similar evolutionary rates High-land = Lowland | $\sigma^2$ | Species retain ancestral ecological traits over time; constraint in physiological evolution limits divergence regardless of elevation. | Bennet et al., (2021) | Species at both elevations will show similar $CT_{min}$ evolution, as all lineages are equally constrained. |   |
| | $\alpha$ | Conserved selective pressures maintain similar strength of stabilizing selection across elevations. | Beaulieu et al., (2012) | No difference in $\alpha$; selective pull toward the $CT_{min}$ optimum is equally strong in both habitat types. |   |
| | $\theta$ | Niche conservatism limits physiological adaptation; highland and lowland species share ancestral $CT_{min}$ optima. | Wiens et al., (2010) | No difference in $\theta$; lowland and highland lineages converge on the same optimal $CT_{min}$ value. |   |
| Higher evolutionary rates in highland species High-land > Lowland | $\sigma^2$ | Highland species have experienced repeated climate oscillations, potentially accelerating evolution as they adapt to shifting conditions; evolutionary lability. | Salazar et al., (2019) | Highland species, facing greater environmental instability, should show faster rates of $CT_{min}$ evolution. |   |
| | $\alpha$ | $CT_{min}$ is experienced at night when behavioral thermoregulation is ineffective, meaning the Bogert effect cannot buffer cold tolerance evolution; highland oscillations further intensify selection. | Muñoz, (2022) | Highland species should show stronger stabilizing selection (higher $\alpha$) on $CT_{min}$, as behavioral buffering is absent at night and climate shifts are more frequent. |   |
| | $\theta$ | Colder highland environments impose lower thermal limits; Bio 6 confirms highland species experience lower minimum temperatures, predicting lower optimal $CT_{min}$. | Sinervo et al., (2010) | Highland species should show lower optimal $CT_{min}$ (lower $\theta$), reflecting colder thermal environments. |   |

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**Table 1.** Continued

| Hypotheses | Parameter | Mechanism | References | Predictions | Expected relations |
|---|---|---|---|---|---|
| Higher evolutionary rates in lowland species Lowland > Highland | $\sigma^2$ | Lowland species experience higher competition, predation, and parasitism pressures, potentially driving rapid adaptive divergence. | Schemske et al., (2009) | Species in lowland environments will exhibit faster rates of $CT_{min}$ evolution, due to greater biotic pressure. |   |
| | $\alpha$ | Diffuse selection from multiple competing biotic pressures in lowlands may relax stabilizing selection on $CT_{min}$, allowing greater drift away from the optimum. | Muñoz, (2022); Wiens et al., (2010) | Lowland species should show weaker stabilizing selection (lower $\alpha$) on $CT_{min}$, as competing selective agents dilute thermal constraint. |   |
| | $\theta$ | Greater biotic pressure in lowlands may select for higher $CT_{min}$ optima, allowing more activity at warmer minimum temperatures to compete and evade predators. | Schemske et al., (2009) | Lowland species should show higher optimal $CT_{min}$ (higher $\theta$), reflecting warmer thermal environments and elevated activity requirements. |   |

For $CT_{min}$, we were unable to run the analysis using the 2,100 m threshold because the final dataset did not include any highland species. We ended up with 24 highland and 40 lowland species at the 700 m criterion, 33 highland and 31 lowland species at 300 m, and 7 highland and 57 lowland species at 1,400 m. For both minimal thermal niche ($_{min}$TN) and $CT_{min}$, we ran all analyses shown below across all elevational thresholds, as data permitted.

## Phylogenetic comparative analyses

For comparative phylogenetic analyses, we analyzed all species together across our two datasets for $_{min}$TN and $CT_{min}$. To estimate ancestral character states for the discrete trait of highland vs. lowland, we used the *fitDiscrete()* function in the "*geiger*" R package (Harmon et al., 2008; R Core Team, 2023) to generate a stochastic character mapping and compared the ER (equal-rates) and ARD (all-rate-different) models using Akaike Information Criterion (AIC) values (Burnham & Anderson, 2002; Sugihara,1978).

We fit 10 evolutionary models (see Supplementary Material 1 and Table S3 for further explanation) to test whether the rate of evolution in thermal traits is associated with biogeographical transitions from lowlands to highlands and vice versa, using the *hOUwie()* function in the "*OUwie*" package in R (Beaulieu and O'Meara, 2012; Boyko et al., 2023). The *hOUwie()* function models the joint evolution of a discrete trait (biogeographical transitions) and a continuous trait ($_{min}$TN or $CT_{min}$). This function integrates the likelihood of the $_{min}$TN and $CT_{min}$ under a generalized Ornstein–Uhlenbeck (OU) process with the likelihood of a biogeographical transition modeled by a hidden Markov process, both influenced by a shared underlying state (lowland or highland; Boyko et al., 2023). These 10 models contain three parameters we deemed important: the rate of stochastic trait evolution ($\sigma^2$), the optimum evolutionary trait value ($\Theta$), and the rate of evolution toward the optimal state ($\alpha$; also called evolutionary pull); the different models allowed both rates and optimal trait value parameters to vary between lowland and highland species (see Supplementary Material 1 for further explanation). In addition, we compared the joint evolution of $_{min}$TN and biogeographical transitions for island and mainland species independently; for this, we compared both groups by using the same 10 *hOUwie* models we utilized for the whole $_{min}$TN dataset (Supplementary Material 1). For $CT_{min}$, since we have 64 species in our dataset, we fitted six of the original 10 *hOUwie* models (Supplementary Material 1, Table S3) to avoid overfitting. Additionally, we compared the joint evolution of $CT_{min}$ and biogeographical transitions for island and mainland species using the same six *hOUwie* models that we run for the cold tolerance data (Supplementary Material 1, Table S3).

We used model averaging within *hOUwie* to test whether biogeographical transitions are associated with distinct evolutionary dynamics in the minimal thermal niche ($_{min}$TN) and cold tolerance ($CT_{min}$; Boyko et al., 2023). Model averaging summarizes information from all models in the set (10 models in total for $_{min}$TN and six for $CT_{min}$), assigning weights to each based on their explanatory strength, as de-

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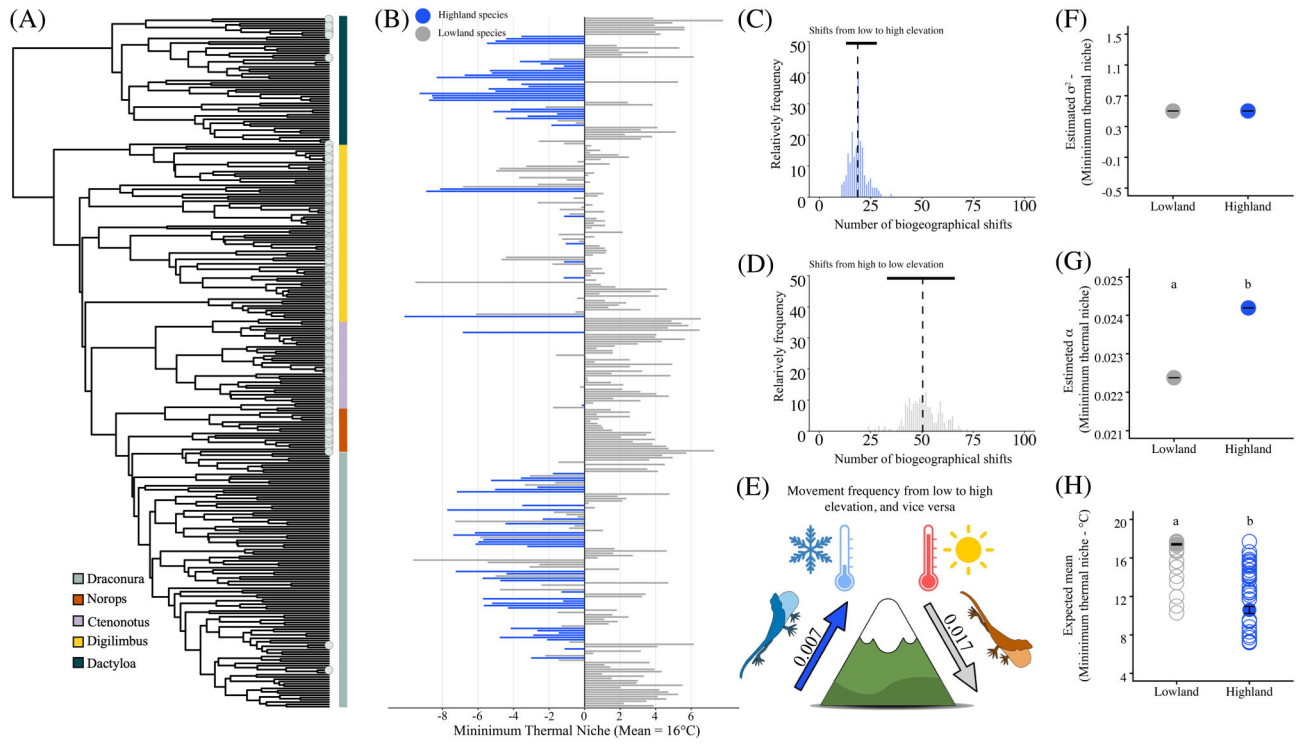



**Figure 1.** Minimum thermal niche ($_{min}$TN) ancestral state reconstruction and *hOUwie* results for the 700 m criterion. (A) Anole phylogeny; circles at the tips represent island species. (B) Minimum thermal niche ($_{min}$TN) values (relative to overall trait mean) for each species. Number of biogeographical gains (C) and losses (D) of high-elevation occupation among 200 stochastic map reconstructions; the vertical dashed line represents the mean value, and the horizontal solid line shows the 95% highest density interval. (E) Transition rate matrix estimated from the ARD model. Illustration by Jesús Bazán Villicaña and Abby Kimball in association with InPrint at Washington University in St. Louis. (F) Evolutionary rate ($\sigma^2$), (G) strength of selection ($\alpha$), and (H) expected trait mean value (°C) estimated for minimum temperature of the coldest month for each elevational group (high- and low-elevation species) estimated from 200 stochastic character maps; the circles represent the mean, and the bars represent the mean standard error. The small letters represent significant differences between the groups.

termined by the sample size-corrected AIC (Akaike, 1998). This method allows us to draw conclusions based on the collective evidence of all models rather than relying solely on the top-performing one (Burnham & Anderson, 2002). Species were categorized as lowland or highland based on an elevation threshold; because no single biologically justified cutoff exists for this system, we applied this approach across a range of thresholds (300, 700, 1,400, and 2,100 m) to assess the sensitivity of our results to this choice. Furthermore, instead of directly comparing parameter estimates ($\sigma^2$, $\alpha$, and $\Theta$), we compared the expected values of the tips, which integrate parameter estimates with the phylogenetic history of each lineage (Beaulieu et al., 2012; Boyko et al., 2023; Butler & King, 2004; Hansen, 1997).

We estimated ancestral geographical states on the phylogeny and 200 histories of stochastic character maps with the *make.simmap()* function in the "*phytools*" package in R (Revell, 2012). Lastly, we quantified phylogenetic signal in the evolution of highland colonization (lowland vs. highland species) by calculating the D statistic in the "*caper*" package in R (Fritz & Purvis, 2010; Orme, 2013) using the function *phylo.d()*. The *D* statistic values range from 0 to 1, where a value of 1 indicates that the observed trait has a low or no phylogenetic signal, and a value of 0 indicates a strong phylogenetic signal. *D* values lower than 0 or higher than 1 suggest exceptionally high conservatism or overdispersion of the observed trait, respectively. We used 10,000 permutations to determine the significance of *D*.

For both $_{min}$TN and $CT_{min}$ data, we discuss our findings for the 700 m criterion in the main text; the results for the other criteria were fairly similar and can be found in the Supplementary Material section.

### Methodological caveats

There are several criteria for defining a mountainous area (e.g., topographic and altitudinal). As mentioned before, we categorized species as lowland or highland based on an elevation threshold (300, 700, 1,400, and 2,100 m); we acknowledge this discretization (i.e., lowland and highland) is a simplification of what is likely a more gradual ecological transition. We used the *hOUwie* framework because it allows us to directly test whether transitions between lowland and highland zones are associated with shifts in the rate and mode of thermal trait evolution, which requires modeling trait evolution as a function of a discrete state, precisely the strength of *hOUwie* for addressing our question.

## Results

We found that anoles have transitioned from lowland to highland habitats 18 times, and from highland to lowland 50 times throughout their evolutionary history (ARD model; Figure 1C and D). However, the transition rates show a directional signal: highland-to-lowland colonization proceeds at roughly 2.4 times the rate of the opposite transition (0.017

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[high to low] vs. 0.007 [low to high]; Figure 1E), suggesting that despite the challenges of warm lowland environments for cold-adapted highland species, the evolutionary pressure to return to lowland conditions is stronger than the pressure to remain in montane habitats. Consistent with this, 70 species representing 18 independent occupations occur above 700 m of elevation, suggesting that highland lineages remain comparatively rare across the anole radiation (70 species out of 303 in our dataset; for results from other cutoffs, see Figures S1–S3, Supplementary Material 2, Table S4).

When we tested whether the transition between lowlands and highlands affects the rate of evolution of minimum thermal niche ($_{min}$TN), we found that the best-fitting *hOUwie* model for $_{min}$TN was the OUMA model (Figure 1; Supplementary Material 1), which fits a single rate of stochastic trait evolution ($\sigma^2 = 0.50$) for lowland and highland species, while allowing different strengths of selection ($\alpha_{lowland} = 0.022$, $\alpha_{highland} = 0.024$) and multiple optimum trait value ($\Theta$; lowland = 17.41°C, highland = 10.59°C) for lowland and highland species (for results for the other elevational thresholds see Supplementary Material 3, Figures S1–S3). When we separated our $_{min}$TN data into island and mainland species, we found that for mainland species the best model OUMV (Supplementary Material 1). This model estimates multiple $\sigma^2$ ($\sigma^2_{lowland} = 0.81$, $\sigma^2_{highland} = 0.33$), and multiple expected means (lowland = 16.11°C, highland = 10.66°C), and a single $\alpha$ (0.033) for both lowland and highland species (Figure 2; for the other elevational criteria, see Supplementary Material 3, Figures S4–S6). For island species the best-fitting *hOUwie* model was OUMA (Figure 2; Supplementary Material 1); this model fit a single $\sigma^2$ for lowland and highland species ($\sigma^2 = 0.00084$), multiple $\alpha$ ($\alpha_{lowland} = 0.50$, $\alpha_{highland} = 0.00030$), and multiple expected means ($\Theta$; lowland = 17.42°C, highland = 17.42°C; the difference between those two was $5.4 \times 10^{-11}$°C) for lowland and highland species (Figure 2; for the other elevational criteria, see Supplementary Material 3, Figures S7–S9).

For most elevational thresholds, the occupation of highland habitats shows strong phylogenetic conservatism. For the 700 m, the $D$ statistic is close to zero ($D = -0.035$) and does not differ significantly from expectations under Brownian motion ($p$-value = .59), while being significantly different from a random distribution ($p$-value < .05; Supplementary Material 4, Table S5). This suggests that highland lineages tend to cluster on the phylogeny more than expected by chance, consistent with the pattern of repeated but phylogenetically non-random colonization of montane environments across the anole radiation.

For $CT_{min}$, the best-fitting *hOUwie* model was BM1. This model assumes similar rates $\sigma^2$ for highland and lowland species (Figure 3; $\sigma^2 = 0.35$). This model also estimates a single $\alpha$ ($\alpha = 0.0000011$) and a single expected $CT_{min}$ value ($\Theta$; 11.40°C) for both lowland and highland species (Figure 3; for results for the other elevational thresholds, see Supplementary Material 3, Figures S10 and S11). Additionally, similar to minimal thermal niche ($_{min}$TN), for $CT_{min}$ the $D$ statistic ($D = 0.40$) falls between random and Brownian motion expectations, being significantly different from a random distribution ($p$-value = .0099), and not different significantly from Brownian motion ($p$-value = .16; Supplementary Material 4, Table S5). This indicates that the distribution of cold tolerance across the phylogeny is more conserved than expected by chance.

When we separated our $CT_{min}$ data between island and mainland species, we found that for island species the best model for the 700 m criterion was BM1, which fit a single expected mean and a single $\alpha$ for lowland and highland species (expected mean value = 9.97°C; $\alpha = 0.00022$). Since we average across all models, $\sigma^2$ estimates reflect model-averaged rates rather than a single best-fit model: $\sigma^2$ (lowland) = 0.30, $\sigma^2$ (highland) = 0.21 (see the "Methods" section). Similarly, for mainland species the best model was BM1, which fit a single expected mean and a single $\alpha$ for lowland and highland species (expected mean value = 9.56°C; $\alpha < 0.00001$); for mainland, we also have separate $\sigma^2$: $\sigma^2$ (lowland) = 0.38, $\sigma^2$ (highland) = 0.21 (Figure 4; for the other elevational criteria, see Supplementary Material 3, Figures S10 and S11).

## Discussion

Occupation of montane habitats from lowland ancestors has occurred at least 18 independent times in the evolutionary history of anoles. Here, we used physiological ($CT_{min}$) and minimal thermal niche ($_{min}$TN) data and applied *hOUwie* to test the joint evolution of a transition biogeographic (a discrete trait) and thermal tolerance traits (a continuous trait: cold tolerance [$CT_{min}$] and minimal thermal niche [$_{min}$TN]). The two traits show complementary but different pictures of how evolution proceeds across elevational zones. For $CT_{min}$, lowland and highland lineages evolve at similar rates ($\sigma^2$), with comparable strength of stabilizing selection ($\alpha$) and similar optimal trait values ($\theta$). This suggests that cold tolerance evolves similarly across montane environments regardless of whether a lineage is lowland or highland, without being pulled toward a fundamentally different physiological optimum. For $_{min}$TN, the pattern is similar, where rates of evolution are comparable across elevations, but highland lineages experience stronger stabilizing selection ($\alpha$) around a lower thermal optimum ($\theta$); this result, while consistent with the temperature-elevation gradient, also reflects tighter macroevolutionary constraint on the climatic niche of montane species. Together, these results suggest that the shift from lowland to highland habitats reorganizes the evolutionary process differently for an organism's physiology compared to its thermal environment. Below, we unpack these findings in more detail.

### Patterns of biogeographical transitions in *Anolis* lizards

Based on the 700 m highland transition definition (Frishkoff et al., 2022), anoles have moved into montane regions at least 18 times and back to lowlands 50 times. Notably, the transition rate into mountainous regions (0.007) is 2.4-fold lower than the reverse transition (0.017; Figure 1). For $CT_{min}$, using the same elevational cutoff (700 m), we observed that there are 43 lowland species and 21 highland species. In the case of the climatic data (minimal thermal niche), *Anolis* species from almost all clades have colonized high-elevation habitats. Additionally, the distribution of these highland occupants exhibits a strong phylogenetic signal ($D = -0.035$; Supplementary Material 4, Table S4). Rather than being randomly dispersed across the tree, these transitions are highly clustered within specific lin-

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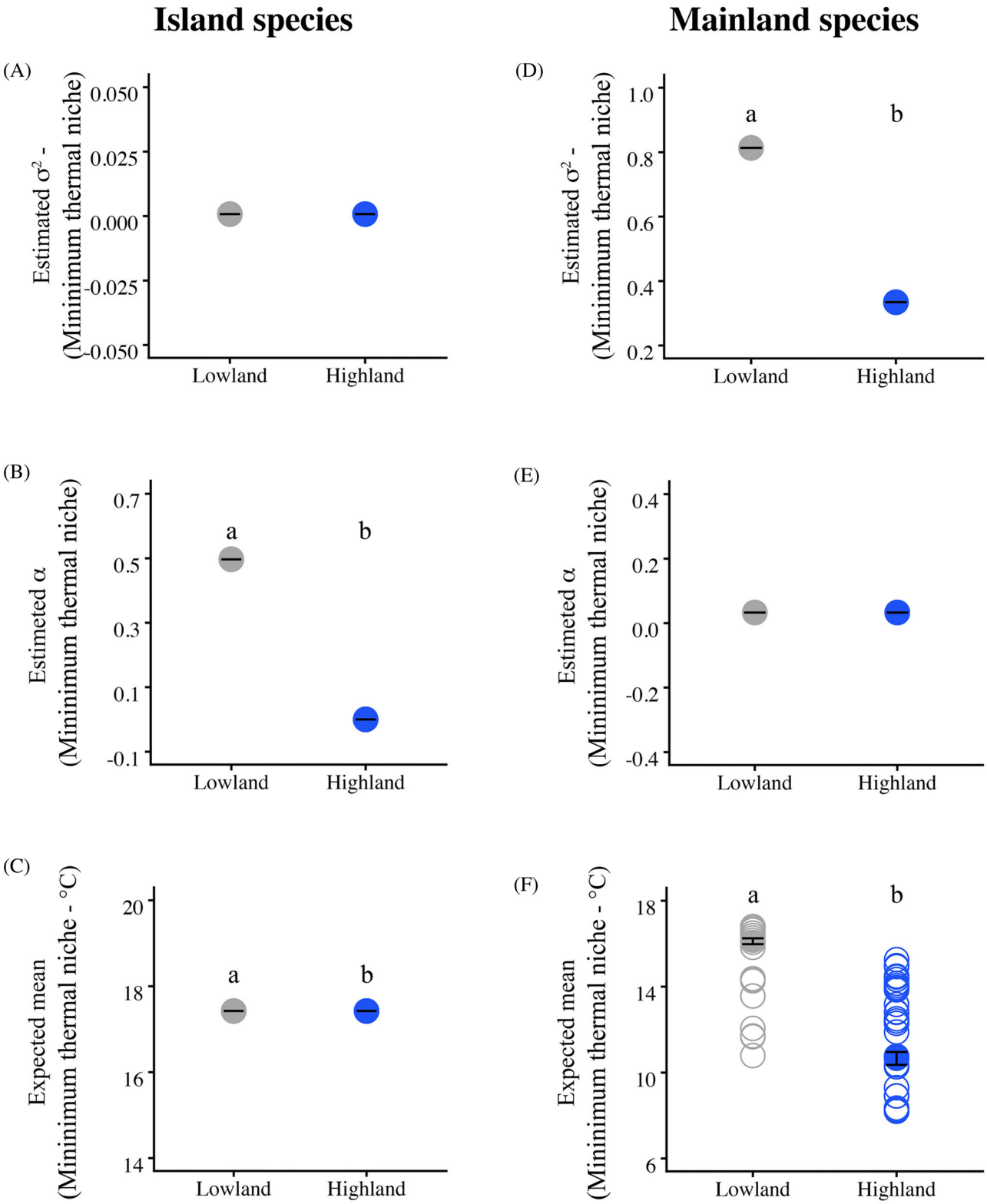



**Figure 2.** *hOUwie* results minimum thermal niche ($_{min}$TN) for island and mainland species. The left column shows results for island species (A–C) and the right column for mainland species (D–F). (A, D) Evolutionary rate ($\sigma^2$), (B, E) strength of selection ($\alpha$), and (C, F) expected trait mean value (°C) for each elevational group (high- and low-elevation species) estimated from 200 stochastic character maps; circles represent the mean, and the bars mean standard error. The small letters represent significant differences between the groups.

eages (Figures 1 and 3). This indicates that while the transition to montane life has occurred multiple times independently across the *Anolis* radiation, these highland lineages also move back to lowland habitats at a comparatively high rate, suggesting that montane occupancy is evolutionarily labile rather than stable. Moreover, only a handful of species

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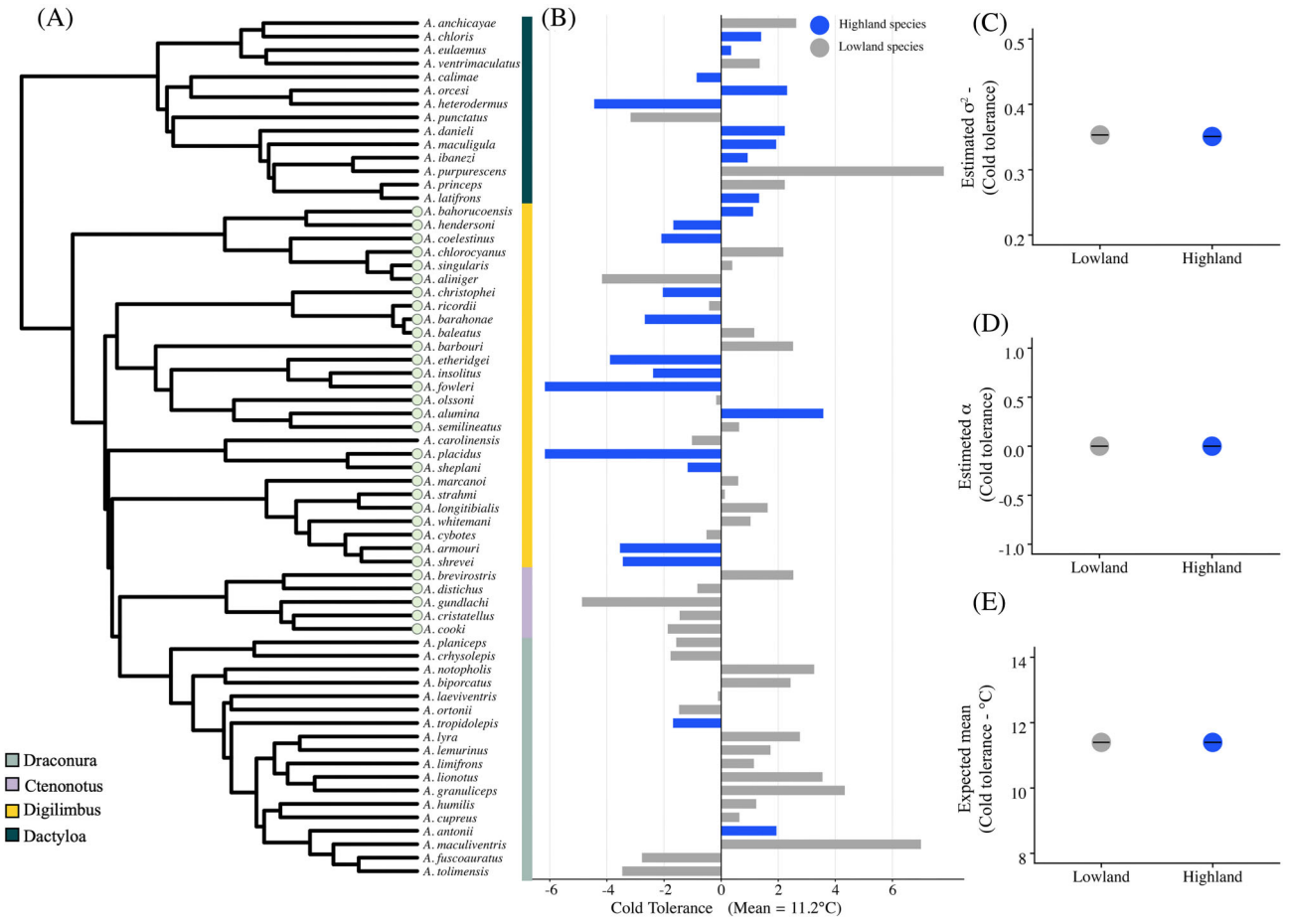



**Figure 3.** Cold tolerance ($CT_{min}$) *hOUwie* results for the 700 m criterion. (A) Anole phylogeny; green circles at the tips represent island species. (B) Cold tolerance values (relative to overall trait mean) for each species. (C) Evolutionary rate ($\sigma^2$), (D) strength of selection ($\alpha$), and (E) expected trait mean value (°C) estimated for cold tolerance for each elevational group (high- and low-elevation species) estimated from 200 stochastic character maps; the circles represent the mean, and the bars represent the mean standard error. The small letters represent significant differences between the groups.

have evolved to occupy habitats at extreme elevations: *Anolis inderenae*, which exhibits the highest minimum elevation (of 2,500 m, and maximum elevation of 3,100 m), and the high-elevation champion, *Anolis heterodermus*, which occurs at elevations as high as 3,750 m and as low as 2,500 (Dunn, 1944; Moreno-Arias & Urbina-Cardona, 2013). For the $CT_{min}$ data, we had data for only three species that evolved to occupy habitats at high elevations in the Andean mountains (>1,400 m), both from the mainland and within the Dactyloa clade: *Anolis orcesi* (found in Ecuador; Figure S11) and *A. calimae* (found in Colombia; Figure S11).

Occupation of colder environments by anoles can occur through a combination of behavioral and physiological mechanisms. During the day, thermoregulatory behavior (i.e., basking) allows lizards to maintain body temperatures at or close to their preferred body temperature even in cooler habitats (Muñoz et al., 2014; Qu & Wiens, 2020). However, at night, when anoles are inactive and cannot bask, physiological adaptations become essential for surviving in colder conditions. As diurnal organisms, anoles lack behavioral strategies to cope with nocturnal cold, making the evolution of cold tolerance critical for high-elevation living. In this context, the ability to function at lower temperatures (e.g., $CT_{min}$) represents an important physiological adaptation. Thus, while behavioral adjustments may play some role (Muñoz et al., 2014), the ability of mainland anoles to occupy higher elevations likely depends primarily on evolutionary changes in cold tolerance ($CT_{min}$), especially because these group species are known to be thermoconformers (Salazar et al., 2019, 2025). Integrating both aspects, climatic niche and thermal physiology, is essential to understanding how anoles diversify across thermal gradients present in the Neotropical montane ranges.

In our study, we combined ecological and physiological data to understand how anole species have evolved to occupy high-elevation environments. By integrating elevational range, climatic variables, and physiological tolerance ($CT_{min}$), we show that highland occupation has occurred repeatedly across the anole radiation but remains relatively rare compared to lowland occupancy. Moreover, the scattered phylogenetic distribution of highland colonizers suggests that the potential for such transitions is not confined to a single lineage but rather emerges multiple times across the anole phylogeny. Together, our findings emphasize the importance of coupling ecological opportunity with physiological adaptation in shaping the diversification of anoles across tropical mountains, providing a framework for understanding how organisms have responded to thermal gradients in the past and how they will respond under ongoing climate change.

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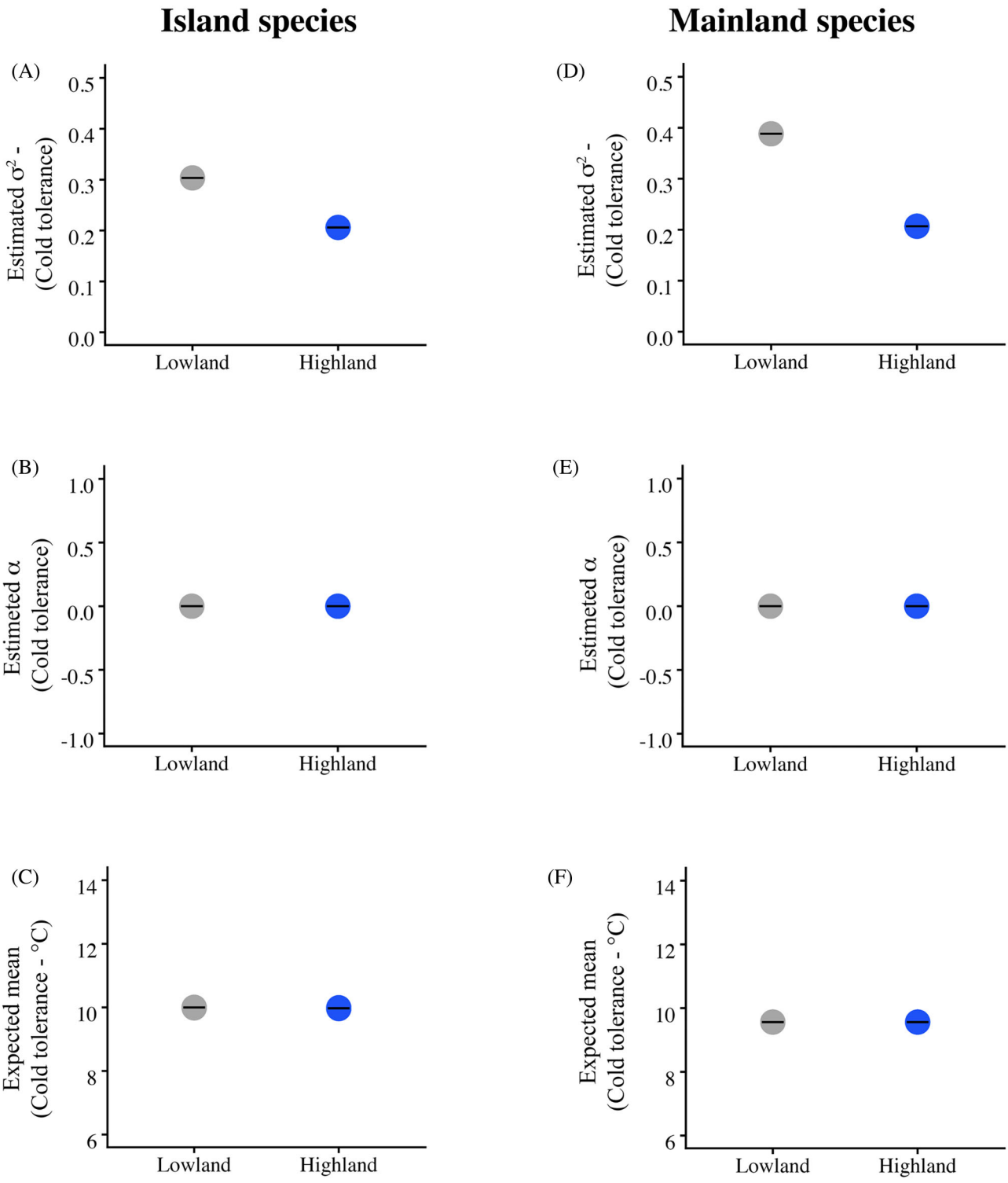



**Figure 4.** *hOUwie* results for cold tolerance ($CT_{min}$) for island and mainland species. The left column shows results for island species (A–C), and the right column for mainland species (D–F). (A, D) Evolutionary rate ($\sigma^2$), (B, E) strength of selection ($\alpha$), and (C, F) expected trait mean value (°C) for each elevational group (high- and low-elevation species) estimated from 200 stochastic character maps; the circles represent the mean, and the bars represent the mean standard error. The small letters represent significant differences between the groups.

## Climatic niche conservatism and selection in highland anoles

For the minimal thermal niche ($_{min}$TN), transitions from lowland to highland were more frequent than the reverse (Figure 1C and D; Table 2), suggesting directional occupation of colder environments. Evolutionary rates ($\sigma^2$) for $_{min}$TN were similar between lowland and highland species (Figure 1F), a pattern consistent with the hypothesis that stronger biotic interactions in lowlands drive faster trait evolution or that climatic instability drives faster trait evolution

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**Table 2.** Number of highland and lowland species and transitions to and from the highlands for each elevational threshold among 200 stochastic maps reconstructions for minimum thermal niche ($_{min}$TN) and cold tolerance ($CT_{min}$).

| Elevational threshold | Number of highland species | Number of transitions to highland | Number of lowland species | Number of reversals to lowland |
|---|---|---|---|---|
| | | Minimum thermal niche | | |
| 700 m criterion | 70 | 30 | 233 | 23 |
| 300 m criterion | 95 | 39 | 208 | 15 |
| 1,400 m criterion | 19 | 10 | 284 | 24 |
| 2,100 m criterion | 1 | 1 | 302 | 0 |
| | | Cold tolerance | | |
| 700 m criterion | 21 | NA | 43 | NA |
| 300 m criterion | 28 | NA | 36 | NA |
| 1,400 m criterion | 7 | NA | 57 | NA |

in highland species (Table 1; Bennet et al., 2021; Rolland et al., 2014; Schemske et al., 2009). What differs between zones is not the tempo of evolution but the nature of the selective regime: the strength of stabilizing selection ($\alpha$) was greater for highland species (Figure 1G), indicating that montane environments impose a tighter evolutionary pull toward the thermal optimum even when overall rates remain comparable. Expected mean $_{min}$TN values were lower in highland species, as expected given that minimum temperatures decline predictably with elevation (Figure 1H). In this regard, the patterns we found for climatic niche evolution can be considered a null hypothesis for understanding how species evolve into montane environments.

Evolutionary trends for $_{min}$TN also differ between mainland and island species. On islands, species face similar evolutionary constraints regardless of elevation ($\sigma^2$; Figure 2B), while on the mainland, evolutionary rates are higher in lowland species, likely reflecting the more intense biotic interactions (e.g., predation, competition) experienced at lower elevations (Schemske et al., 2009); however, we did not test this in our study. Island species overall show higher strength of stabilizing selection compared to mainland species, suggesting more homogeneous selective regimes in insular environments, and converge on higher expected mean Bio 6 (minimum temperature of the coldest month, here measure as $_{min}$TN—minimum thermal niche) values, indicating that mainland and island lineages are evolving toward distinct adaptive peaks in climatic space.

One possible explanation for the frequency of transitions to high elevations in anoles is that colonization of higher elevations may not have been an active dispersal process driven by macroclimate alone; rather, species may have been carried vicariantly as mountains rose. Anoles originated and radiated first in South America during the Paleocene–Eocene, approximately 64.4–46.3 Myr, which predates the orogeny of the Andes (10–3 Myr; Poe et al., 2017). This timeline suggests that high-elevation colonization was partly the result of anoles riding the uplifting Andes, ultimately leading to speciation and diversification (Salazar et al., 2026); comparable patterns have been found in other Andean lizard groups, including Liolaemidae (Ibargüengoytía et al., 2021) and Andean squamates more broadly (Velasco et al., 2026). A similar scenario seems to have occurred in Central America, where anole cladogenesis is correlated with intense tectonic activity during the mid-Miocene (15–10 Myr; Poe et al., 2017). Furthermore, dispersal into high elevations may be relatively straightforward via upslope movement, whereas dispersal within high-elevation zones is often constrained by complex topography (e.g., Moreno-Arias & Méndez-Galeano, 2026; Moreno-Arias et al., 2023; but see Salazar et al., 2026).

## Cold tolerance evolution across biogeographical transitions

Similar to the climatic signal, cold tolerance ($CT_{min}$) shows a comparable evolutionary pattern. Physiological data could give a more informative picture than climatic niche evolution on how species adapt to montane environments, given that physiology is directly shaped by environmental temperature variation (Card et al., 2018); Muñoz et al., 2014; Salazar et al., 2019, 2025). For instance, lowland and highland lineages show comparable rates of $CT_{min}$ evolution ($\sigma^2$; Figure 3C), also with comparable strength of stabilizing selection ($\alpha$) is between groups (Figure 3E), and optimal $CT_{min}$ values ($\theta$; Figure 3E) that do not differ between biogeographical transitions.

This similarity in evolutionary rates ($\sigma^2$) between lowland and highland lineages suggests that physiological constraints limit divergence in cold tolerance regardless of elevational grouping. The low and comparable $\alpha$ estimates in both groups (Figure 3D) further suggest weak stabilizing selection around the $CT_{min}$ optimum in both lowland and highland lineages, rather than strong constraint, consistent with a process closer to Brownian motion (BM1) evolution than to a tightly bounded adaptive peak (OU process); the shared $CT_{min}$ optimum across groups may nonetheless reflect a lower physiological boundary in thermal tolerance, probably because of cold night temperatures. This interpretation is consistent with the *D*-statistic results ($D = 0.40$), which indicate that the phylogenetic signal for $CT_{min}$ follows Brownian motion expectations, a pattern that aligns with the near-zero $\alpha$ estimates, because weak stabilizing selection would be expected to produce dynamics closer to Brownian motion than to a tightly constrained OU process.

One possible explanation for this pattern is behavioral thermoregulation buffering physiological traits from direct selection (the *Bogert effect*): If lineages can behaviorally compensate for thermal variation across elevations (via microhabitat selection, activity timing, or basking behavior), $CT_{min}$ itself may experience relaxed and similar selective pressure across elevational contexts, even as macroclimatic conditions diverge (Muñoz et al., 2014; Muñoz, & Losos, 2018; Salazar et al., 2019, 2025).

For cold tolerance, mainland and island lineages follow the same evolutionary model (BM1; Figure 4), indicating

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that the mode of cold tolerance evolution does not differ between these two biogeographic regions (Figure 4). However, in contrast to previous studies (Salazar et al., 2019), we did not find a consistent pattern of mainland species showing higher expected $CT_{min}$, stronger stabilizing selection, or faster evolutionary rates than island species. Instead, expected $CT_{min}$ was slightly higher in island (9.97°C) than mainland lineages (9.56°C), and $\alpha$ was higher in island ($\alpha = 0.00022$) than mainland lineages ($\alpha < 0.00001$), suggesting comparatively stronger stabilizing selection on cold tolerance in island populations. This contrasts with our previous study (Salazar et al., 2019), though the dataset in that study was limited to 31 species with cold tolerance data, and the broader taxonomic sampling herein may better capture the range of variation in mainland–island cold tolerance evolution. Evolutionary rates showed a more mixed pattern: lowland mainland populations evolved somewhat faster than their island counterparts ($\sigma^2 = 0.38$ vs. 0.30), whereas highland populations showed nearly identical rates in both regions ($\sigma^2 = 0.21$). Because these $\sigma^2$ values are model-averaged across the full candidate model set rather than derived from a single best-fit model, we interpret them with some caution (see the "Methods" section). Together, these results suggest that although mainland and island lineages share a common mode of cold tolerance evolution, the tempo of evolution and specific parameter estimates do not follow a simple mainland-faster/island-slower pattern, and the ecological mechanisms often invoked to explain mainland–island differences in trait evolution (e.g., thermal heterogeneity, biotic interactions; Caetano & Harmon, 2019; Losos & Ricklefs, 2009; Sunday et al., 2011) do not straightforwardly account for the direction of the pattern we observe here.

Interestingly, cold adaptation in anoles is not exclusively a montane phenomenon. *Anolis carolinensis* (Digilimbus clade) and *A. nebulosus* (Draconura clade) occupy environments that can reach temperatures as cold as those experienced by Andean highland anoles such as *A. inderenae* and *A. heterodermus*, demonstrating that cold tolerance has also evolved in response to latitudinal thermal gradients. This parallel is expected given that an increase in latitude of approximately 880 km is thermally equivalent to a 1,000 m increase in elevation (Urry et al., 2017). Why anoles have not been able to colonize elevations higher than 3,750 m (Moreno-Arias & Urbina-Cardona, 2013) or latitudes greater than 36° likely reflects hard physiological constraints on cold tolerance, though the precise nature of these limits remains to be determined.

## Decoupling of thermal environment and physiology in montane colonization

Taken together, the $_{min}$TN and $CT_{min}$ results show that thermal environment and cold tolerance respond differently to the same biogeographic transition and that the shift from lowland to highland habitats reorganizes the evolutionary process in trait-specific ways. For $_{min}$TN, evolutionary rates ($\sigma^2$) remain comparable between lowland and highland lineages, but both the strength of stabilizing selection ($\alpha$) and the optimal trait value ($\theta$) differ significantly, with highland lineages showing stronger selection around a substantially colder optimum (~11°C vs. ~17°C). This pattern reflects the physical reality that lower limits in thermal niches decline with elevation, coupled with tighter, more strongly enforced climatic niche conservatism in montane environments. For $CT_{min}$, by contrast, none of the three parameters ($\sigma^2$, $\alpha$, or $\theta$) differ appreciably between elevational zones, despite highland lineages occupying this same colder thermal environment. This indicates that physiological cold tolerance does not track the climatic niche shift the way minimal thermal niche ($_{min}$TN) does, revealing a decoupling between the climatic niche and thermal physiology. Whether this reflects true evolutionary stasis in cold tolerance optima or a limitation of current sampling (given that only a handful of species occupy extreme elevations in our $CT_{min}$ dataset) warrants further investigation as broader physiological datasets become available (see, e.g., Gouveia et al. 2013; Qu & Wiens, 2020).

Additionally, this contrast between traits also has broader implications for how we interpret macroevolutionary patterns across elevational gradients. The $_{min}$TN signal, while informative about niche dynamics, is partly shaped by the physical reality that atmospheric temperatures decline with elevation, making some degree of differentiation between lowland and highland lineages an expected geographic outcome rather than an independent evolutionary result. The cold tolerance patterns, by contrast, reflect a laboratory-derived measure of physiological performance (critical thermal limits; $CT_{min}$ and $CT_{max}$ [heat tolerance]) rather than a direct field measurement of thermal experience, such as body temperature. Although $CT_{min}$ is not measured under natural conditions, it captures an intrinsic physiological trait that is more directly tied to selection on thermal tolerance itself and is therefore informative as an evolutionary signal in a different way than climatic-niche variables. That these two layers of thermal biology—one distributional-demographical ($_{min}$TN) and one physiological ($CT_{min}$)—tell different stories emphasizes the importance of combining environmental and physiological data when assessing how species adapt across elevational gradients. Moreover, despite the higher rate of highland-to-lowland transition within the anole radiation, the comparatively frequent instances of lowland-to-highland colonization we recovered contrast with broader vertebrate diversity patterns, where highland colonization itself is relatively rare (e.g., amphibians, birds, and mammals; Antonelli et al., 2018), possibly reflecting the high evolutionary lability of cold tolerance across the anole radiation (Muñoz et al., 2014; Salazar et al., 2019).

## Future directions in cold adaptation and climate change resilience

In recent years, studies have focused on whether species will be able to adapt to warming temperatures and what nonevolutionary responses they might have to those new temperatures (e.g., Muñoz et al., 2014; Pörtner et al., 2001; von May et al., 2017; Wood & MacDonald, 1997). Although cold adaptation has been widely studied across animals, including tropical *Anolis* (e.g., Campbell-Staton et al., 2017; Giska et al., 2021; Stroud et al., 2020), research remains geographically biased. In particular, the tropical Andean mountains have received comparatively little attention (but see Bota-Sierra et al., 2022; Salazar et al., 2019, 2025; Velasco et al., 2026), despite their steep thermal gradients and high species diversity. To understand how ancestrally tropical, warm-adapted species have been able to adapt to

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the cold conditions of those mountains, future investigation of thermal physiology—cold tolerance in particular—of species that inhabit multiple mountain ranges would be useful. In particular, future questions should address the possible factors that have influenced the evolution and transition of tropical species from low-elevation areas to high-elevation environments (e.g., Montoya-Cruz et al. 2024 ). In addition, understanding the phylogeography and intraspecific niche evolution of groups with wide altitudinal distributions will help address questions regarding the colonization of mountaintops and how thermal tolerance has evolved within a species or species complexes. Why highland species transition back to lowlands at a high rate is another phenomenon that remains poorly understood (Figure 1E). Answers to these questions may have given us light to understand how species might be able to adapt to a changing climate.

Finally, our study uses a broad-scale framework that treats highland regions as ecologically comparable, although we did test for island-mainland differences, we recognize that this assumption may mask meaningful regional variation. For example, species occurring at similar elevations may experience different ecological and evolutionary dynamics depending on whether they inhabit the Greater or Lesser Antilles versus mainland Central and South America. These differences could shape how anole lineages shift between lowland and highland areas. This level of simplification was necessary to identify general patterns across large spatial and phylogenetic scales, but it also emphasizes the value of studies that account for regional climate histories and lineage-specific traits. Refining how highland systems are compared across taxa and regions will be essential to understanding the evolutionary processes underlying mountain colonization.

## Conclusions

Tropical montane environments present unique challenges for reptiles adapted to warm, lowland conditions, yet *Anolis* lizards have overcome these barriers through multiple independent occupations of high-elevation habitats across Central and South America and the Caribbean. Although highland-to-lowland reversions are more frequent in absolute number (18 transitions vs. 50 reversions), this pattern likely reflects the markedly larger pool of lowland species in the phylogeny rather than a true directional bias in transition rates. Because raw transition counts are confounded by unequal species richness between zones, we rely on the estimated transition rates rather than absolute counts to assess directional bias. The transition rates show a clearer signal: The rate of colonization into montane regions (0.007) is roughly 2.4-fold lower than the reverse transition out of the highlands (0.017; Figure 1), indicating that once lineages establish in lowland environments, colonization of montane habitats is comparatively rare, while highland lineages return to lowland conditions at a substantially higher rate. Interestingly, the extreme elevational ranges achieved by species such as *Anolis heterodermus* and *A. inderenae* suggest a complex interplay between anole evolutionary history and the tectonic orogeny of major mountain ranges, where passive vicariance and active colonization are difficult to disentangle.

Lastly, our results show that thermal environment and physiological cold tolerance evolve along fundamentally different trajectories across the same biogeographic transition. While $_{min}$TN tracks elevation through shifts in selective strength and optimal trait value (an expected geographic outcome), $CT_{min}$ remains evolutionarily stable across zones, suggesting physiology is decoupled from the climatic gradient it inhabits. This decoupling emphasizes that environmental and physiological data capture distinct, complementary layers of thermal biology. Therefore, relying solely on climatic niche data provides only a geographical perspective on how species occupy high elevations and lacks an organismic view of how species adapt in a fine-grained way to montane environments. Whether the absence of detectable divergence in evolutionary rates, strength of evolution, and cold tolerance optima between lowland and highland species reflects a genuine evolutionary constraint or a sampling limitation remains an open question, and one that more complete physiological datasets will be able to address.

## Supplementary material

Supplementary material is available online at *Evolution*.

## Data availability

The data we used is included in the supplementary material, and the codes and the phylogenetic tree that support our analyses are going to be available in Zenodo: https://doi.org/10.5281/zenodo.22235975.

## Author contributions

Jhan C. Salazar developed the original idea, analyzed the data, and contributed to the manuscript draft. Adam C. Algar gathered the data and contributed to the manuscript draft. Steven Poe gathered the data and contributed to the manuscript draft. Jonathan B. Losos developed the original idea and contributed to the manuscript draft. Julián A. Velasco gathered the data, developed the original idea, analyzed the data, and contributed to the manuscript draft. All authors contributed to the discussion of the results and provided substantial support to the manuscript development. Julián A. Velasco and Jonathan B. Losos supervised the work.

## Funding

This research was made possible by funding from the Human Frontier of Science Program (HFSP—RGP0030/2020; to Jonathan B. Losos) and from the Dirección General de Asustos del Personal Académico/Programa de Apoyo a Proyectos de Investigación e Innovación Tecnológica (DGAPA/PAPIIT—IA206523; to Julián A. Velasco).

## Conflict of interest

We declare no competing interests.

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## Acknowledgments

We acknowledge Donald B. Miles for his constructive critique and feedback on the manuscript. We want to thank James Boyko for providing us with help with the hOUwie package. We also want to thank Rafael J. Feliciano and Luis A. Chica for their help in troubleshooting our R script and helping us run our analysis on their supercomputers.

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Received July 10, 2024; revisions received August 6, 2026; accepted August 11, 2026
Associate Editor: Diogo Provete; Handling Editor: Hélène Morlon
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